



Electromagnetic Field Mapping

Numerical Simulation of Electric Dipole Potential and Field Distribution

Abstract

This report details the computational modeling and electrostatic visualization of a two-dimensional electric dipole. By leveraging Coulomb's Law and the Principle of Superposition, the spatial electric potential $V(x, y)$ and vector electric field $\mathbf{E}(x, y)$ are calculated numerically in MATLAB to illustrate fundamental electrostatic behaviors around point charges.

1 Theoretical Background

Consider an electric dipole consisting of two equal and opposite point charges, $+q$ and $-q$, separated by a distance $d = 20$ units along the x -axis in free space ($\epsilon_r = 1$).

According to Coulomb's Law, the electric potential V and electric field intensity $\mathbf{E}$ generated by a single point charge at a displacement vector $\mathbf{r}$ are given by:

$$V = \frac{1}{4\pi\epsilon_0\epsilon_r} \frac{q}{r} \quad (1)$$

$$\mathbf{E} = \frac{1}{4\pi\epsilon_0\epsilon_r} \frac{q}{r^2} \hat{\mathbf{a}}_r = \frac{1}{4\pi\epsilon_0\epsilon_r} \frac{q}{r^3} \mathbf{r} \quad (2)$$

By applying the **Principle of Superposition**, the total potential and field at any arbitrary point (x, y) in the spatial domain are the vector and scalar sums of the individual contributions from both charges:

$$V_{\text{total}}(x, y) = V_1 + V_2 = \frac{q}{4\pi\epsilon_0} \left(\frac{1}{r_1} - \frac{1}{r_2} \right) \quad (3)$$

$$\mathbf{E}_{\text{total}}(x, y) = \mathbf{E}_1 + \mathbf{E}_2 \quad (4)$$

where $r_1 = \sqrt{(x - 10)^2 + y^2}$ and $r_2 = \sqrt{(x + 10)^2 + y^2}$ represent distances from $+q$ and $-q$, respectively.

2 Computational Implementation

A discrete 101×101 spatial mesh grid is defined in MATLAB. To clearly illustrate field directions regardless of proximity to the charge singularities, normalized vector direction components are evaluated for visualization via vector field (quiver) plots and logarithmic magnitude contours.

```
1 % Total Field Components via Superposition Principle
2 Ex = Ex1 + Ex2;
3 Ey = Ey1 + Ey2;
4 V = V1 + V2;
5
6 % Vector Direction Normalization
```



```
7      E_mag = sqrt(Ex.^2 + Ey.^2);  
8      Ex_norm = Ex ./ E_mag;  
9      Ey_norm = Ey ./ E_mag;  
10
```

Listing 1: Core Superposition Logic in MATLAB

The resulting visualizations demonstrate the characteristic dipole field configuration, highlighting the potential well around $-q$ and peak potential around $+q$.